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26 March 1984

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19 October 1983
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31 May 1967

MILITARY STANDARD

FASTENER TEST METHODS

FSC 53GP

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DEPARTMENT OF DEFENSE
Washington, DC 20301

FASTENER TEST METHODS

DOD-STD-1312A

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specification and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

1. Background. The selection of fasteners in the design and construction phase of weapons systems and equipments and the analysis of fasteners used therein for parameters, such as suitability, standardization, and simplification requires the evaluation and correlation of test data and criteria. In evaluating these functions, it has become increasingly evident that the data and criteria received from prime contractors, fastener producers, private laboratories, military documents, and military laboratories cannot be adequately correlated. The chief reason for this is the absence of unified standards prescribing test methods, procedures, fixtures, data analysis, and data presentation. This disparity encourages and often requires duplication of testing and reduces reliability of weapons systems and equipments, thereby impairing operations involving engineering control, design maintenance, quality assurance, standardization, and logistics simplification. To correct this situation it was established that a unified set of test methods and procedures is required to serve as a basic engineering tool for obtaining and presenting engineering data. As a result, this document was developed and prepared to satisfy this vital need. This standard delineates standard test methods, procedures, conditions, fixtures, and methods of recording, analyzing and presenting test data. The purpose of the resulting test data is to determine the capability of fasteners to withstand the various environmental and mechanical conditions encountered when used in military weapons systems.

2. Benefits. Many benefits are to be expected by unified standard tests for fasteners. The following is a list of benefits that can be expected to accrue from using this standard.

a. Availability of testing techniques and criteria necessary for the design and development of increasingly superior structural fasteners. In the past, when a new fastener was needed to solve a problem, the necessary test procedures had to be developed and written; also, the necessary fixtures had to be designed. This always imposed delays, additional expenses, variations in test methods and in test results.

b. Increasing incidences of correlation among fastener test data and design engineering techniques and design criteria for weapons systems and equipments. As new systems are designed to achieve higher performance for a given volume and mass of material, it is essential that design data for structural joints (a large portion of which is obtained by testing fasteners) maintain a close correlation with engineering techniques and design criteria. Where weight is of importance, such as in aircraft structures, such close correlation will provide a substantial reduction in weight.

c. Availability of testing techniques and criteria necessary to maintain consistent quality in the procurement of fasteners.

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d. Reduction in the time and effort required for the development of procurement specifications for fasteners by having a series of standard tests available.

e. Elimination of duplication in the development of test methods and documents by individual fastener producers, Government contractors, and military activities.

f. Elimination of duplication of effort in research and evaluation due to lack of unified standard test methods. When test methods are not standard, results cannot be correlated; therefore, producers, prime contractors, and military activities will not accept each other's results and will duplicate research, development, and evaluation work.

g. For logistics simplification, fastener attributes must be reduced to a common denominator so that variation in type can be compared. Data and the presentation thereof, by means of this standard, will be provided in such form that various types and variations in types of fasteners may be compared to determine their degree of interchangeability, substitutability, and replaceability.

h. Time and effort required to reach agreement in regard to specific engineering changes will be reduced by providing unified standard tests for use in evaluating new fasteners.

i. This standard will provide a means for establishing guidelines for evaluating fastener test data and engineering techniques.

j. Industry and Government design personnel will become thoroughly indoctrinated in the use of a single standard prescribing unified standard test methods.

ACKNOWLEDGEMENT

This standard was prepared by the Fastener Testing Development Group (FTDG), consisting of representatives of Government agencies, engineering societies, trade associations, independent testing laboratories, the using industry, and the fastener producing industries.

With the increased implementation of the International System of Units (SI) or metric system in military specifications and standards, the Department of Defense (DoD) has approved its usage in this standard. Therefore, this issue, prepared under DoD Project Number 53GP-0142-20 by the FTDG, includes test methods in the SI and inch-pound (I-P) units.

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1. SCOPE

1.1 Applicability. This standard establishes methods for testing fasteners, covering both the SI and I-P units, to determine their physical and mechanical properties. These standard test methods will yield reproducible data for use in research, development, procurement or product application. These test methods shall be observed when specifically referenced in the applicable documents. In case of conflict between the provisions of these methods and the individual test procedure in the specification for a particular item, the latter shall take precedence. This document exists as a dynamic standard subject to addition and change as reflected by the "state-of-the-art."

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-962	Outline of Forms and Instructions for the Preparation of Military Standards and Military Handbooks

2.1.2 Documents applicable to a particular test method bookform standard part or section shall be listed therein.

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. DEFINITIONS

3.1 SI - The International System of Units (from the French "Le Systeme International d'Unites"), as modified for use in the United States by the Secretary of Commerce.

Inch-pound units - Units based upon the yard and the pound commonly used in the United States of America and defined by the National Bureau of Standards. Note that units having the same names in other countries may differ in magnitude.

3.2 Definitions applicable to a particular test method bookform standard part or section shall be listed therein.

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4. GENERAL REQUIREMENTS

4.1 Test method numbering system. This standard consists of a general basic document and supporting parts or sections. Each part addresses a specific test method which is designated by a dash number commencing with number 1 for the I-P system of measurement and number 101 for the SI system of measurement. A different dash number is assigned to each test method.

4.2 Document format structure. Each test method is prepared as an individual document in the format of a bookform standard in accordance with the guidelines of MIL-STD-962. The individual bookform standard part is identified by the basic document identifier, MIL-STD-1312 for the I-P system of measurement and DOD-STD-1312 for the SI system of measurement, followed by a sequential dash number. The dash number is identical to the number of the test method. Examples:

I-P System

MIL-STD-1312-1

- Dash number
  Test Method No. 1, Salt Spray
- Document identifier assigned to the basic standard

SI System

DOD-STD-1312-101

- Dash number
  Test Method No. 101, Salt Spray
- Document identifier assigned to the basic standard

4.3 Method of revision. The individual bookform standard part is revised and issued independent of the other parts. Revisions are identified by a capital Gothic letter immediately following the document identifier dash number. Examples:

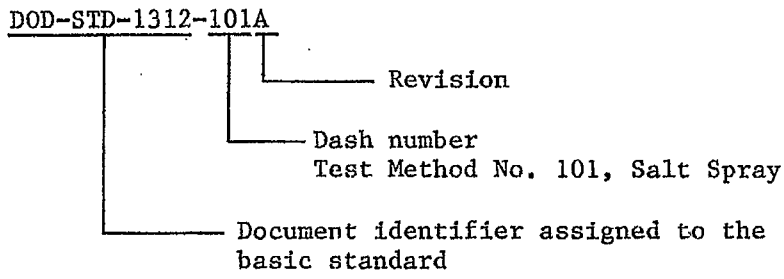
I-P System

MIL-STD-1312-1A

- Revision
- Dash number
  Test Method No. 1, Salt Spray
- Document identifier assigned to the basic standard

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SI System



Changes to individual pages shall be accomplished in accordance with the change notice guidelines of MIL-STD-962.

4.3.1 Revision to method number. Revision to the document will also reflect a change to the test method number. The revision to the test method shall be denoted by a letter following the original number. For example: The original test method number assigned "Test Method 1, Salt Spray" shall be changed to "Test Method 1A, Salt Spray" or "Test Method 101, Salt Spray" shall be changed to "Test Method 101A, Salt Spray" when revised.

4.4 Selection of tests. The tests contained herein are intended to be used to provide uniform testing methods and presentation of data for fasteners and fastening. These tests shall be applicable for research, development and procurement of weapons systems, equipments, and fasteners. In selecting the tests that are to be applicable to fasteners in a specific category or type, it is required that those fastener properties be determined that are needed for incorporation of specific fasteners in specific types of weapons systems and equipments and those properties that are essential for fabrication of a satisfactory fastener. After these essential properties have been determined, the appropriate test shall be selected and specified for each property. Care shall be taken to specify only those tests that are needed.

4.4.1 The test methods available for reference and selection are outlined in the following Appendices:

a. APPENDIX A - List of available test methods in the I-P system of measurement.

b. APPENDIX B - List of available test methods in the SI system of measurement.

4.5 Method of reference. This standard shall be referenced in general and detailed specifications or other documentation for military weapons systems. Tests shall be referenced in the individual fastener specification, fastener specification control drawing, etc., by specifying:

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a. The standard number without the revision letter.

b. The standard number with the applicable dash number and revision letter. For example: "the fastener shall be tested for stress durability in accordance with MIL-STD-1312-5A."

c. Test criteria or property values required.

5. DETAIL REQUIREMENTS

5.1 Detail requirements shall be incorporated in the individual test method bookform standard part or section.

Custodians:
Army - AR
Navy - AS
Air Force - 11

Reviewer activities:
Army - AV, AR
Navy - AS, SH
Air Force - 11
DLA - IS

Preparing activity:
Navy - AS
(Project No. 53GP-0142-20)

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APPENDIX A

LIST OF AVAILABLE TEST METHODS IN I-P SYSTEM OF MEASUREMENT

10. Test methods. The following test methods in the I-P system are available for selection and reference:

MIL-STD-1312-1	Salt Spray
MIL-STD-1312-2	Interaction
MIL-STD-1312-3	Humidity
MIL-STD-1312-4	Lap Joint Shear
MIL-STD-1312-5	Stress Durability
MIL-STD-1312-6	Hardness
MIL-STD-1312-7	Vibration
MIL-STD-1312-8	Tensile Strength
MIL-STD-1312-9	Stress Corrosion
MIL-STD-1312-10	Stress Rupture
MIL-STD-1312-11	Tension Fatigue
MIL-STD-1312-12	Thickness of Metallic Coatings
MIL-STD-1312-13	Double Shear
MIL-STD-1312-14	Stress Durability (Internally Threaded Fasteners)
MIL-STD-1312-15	Torque-Tension
MIL-STD-1312-16	Clamping Force for Installation Formed Fasteners
MIL-STD-1312-17	Stress Relaxation
MIL-STD-1312-18	Elevated Temperature Tensile Strength
MIL-STD-1312-19	Fastener Sealing
MIL-STD-1312-20	Single Shear
MIL-STD-1312-21	Shear Joint Fatigue, Constant Amplitude
MIL-STD-1312-22	Receptacle Push-out, Panel Fasteners
MIL-STD-1312-23	Tensile Strength of Panel Fasteners
MIL-STD-1312-24	Receptacle Torque-out, Panel Fasteners
MIL-STD-1312-25	Driving Recess Torque (Quality Conformance Test)
MIL-STD-1312-26	Structural Panel Fastener Lap Joint Shear
MIL-STD-1312-27	Panel Fastener Sheet Pull-up
MIL-STD-1312-28	Elevated Temperature Double Shear
MIL-STD-1312-29	Shank Expanding
MIL-STD-1312-30	Sheet Pull-up of Blind Fasteners

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APPENDIX B

LIST OF AVAILABLE TEST METHODS IN SI SYSTEM OF MEASUREMENT

10. Test methods. The following test methods in the SI system are available for selection and reference.

DOD-STD-1312-105	Stress Durability
DOD-STD-1312-107	Vibration
DOD-STD-1312-108	Tensile Strength
DOD-STD-1312-109	Stress Corrosion
DOD-STD-1312-111	Tension Fatigue
DOD-STD-1312-113	Double Shear

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